

POKAYOKE (Error Proofing) Procedure

Process 1	Scope of error proofing	Personnel resource	Cross-Functional Team
		Material resource	No designation
INPUT			
The suppliers shall establish PokaYoke systems for manufacturing processes and operations that correspond to the following items. a) Process or operation that have caused market quality problem or outflow b) Process or operation which is deemed necessary with consideration of the following items along with the result of PFMEA conducted by the supplier: - Possibility of occurrence: frequency of occurrence per error (failure) mode - Degree of criticalness: degree of severity per error mode - Degree of spreading prevention: degree of detection per error mode			
In addition, the suppliers shall consider the following items as basic concepts of PokaYoke:			
Preventive action (Prevention)	● Eliminate factors causing additional work or constraining operations and render such work or caution unnecessary. ● Replace manual operations with other more assured methods (i.e. employ machines and/or equipment). ● Design operations to be easy for operators to perform.		
Mitigation of errors	● Employ measures to preclude subsequent processes from proceeding operations until the cause of error is removed or corrected. ● In order to minimize effects to products when operation error occurs, provide shock-absorbing dunnage, protectors, etc., or arrange operation sequences into parallel.		
The suppliers shall request a specification change where employing error proofing safeguards into product specifications for miss assembly or reverse assembly of parts (see SQM 2.4 Drawing and Specification Change and Application).			
Note			
Concrete examples of PokaYoke (error proofing) applied to processes and operations are as follows: - If operation error occurs, parts will not be mounted on jigs - Machine will not start if an operation error or a problem with parts exists - Correct dispersion in operations or equipment automatically, while proceeding with work - Subsequent process verifies operation results of a preceding process and detect problems - Identify parts, jigs, etc., by color, size and shape to distinguish - Automate work operations - Take measures against parts fall including shock absorption when dropped			
Process 2	Evaluating the effectiveness of error proofing	Personnel resource	Cross-Functional Team
		Material resource	No designation
INPUT			
Before applying PokaYoke (error proofing) to process or operation, the suppliers shall evaluate the effectiveness of PokaYoke by considering the following items. a) Capability of detecting operation errors, problems with parts, etc. b) Detection capability of difference in parts location and/or arrangement c) Functions to prevent easy change and unwanted operation of the equipment d) Possibility of the error proofing function to damage parts			
Process 3	Maintaining error proofing effectiveness	Personnel resource	Cross-Functional Team
		Material resource	No designation
INPUT			
The suppliers shall periodically evaluate the effectiveness of the PokaYoke (error proofing) applied to process and operation, based on information and data that include the following items. a) Ratio of non-conforming products detected b) Quantity and details of non-conforming products that were not detected and flowed out into subsequent process c) Operating ratio and failure ratio of error proofing d) Increase or decrease in costs, labor, etc. for error proofing			
Process 4	Feedback from error proofing	Personnel resource	Cross-Functional Team
		Material resource	No designation
INPUT			
The suppliers shall provide feedback on the following items to the responsible party for nonconformities detected in PokaYoke (error proofing), and shall use the feedback to take measures to containment and root cause elimination. a) Non-conforming products detected by error proofing b) Quantity and details of detected non-conforming products c) Corrective action taken against the problem detected d) Specification change request (see SQM 2.4 Drawing and Specification Change and Application)			